

EBN111 CLASS TEST 2 / KLASTOETS 2

8 April 2016

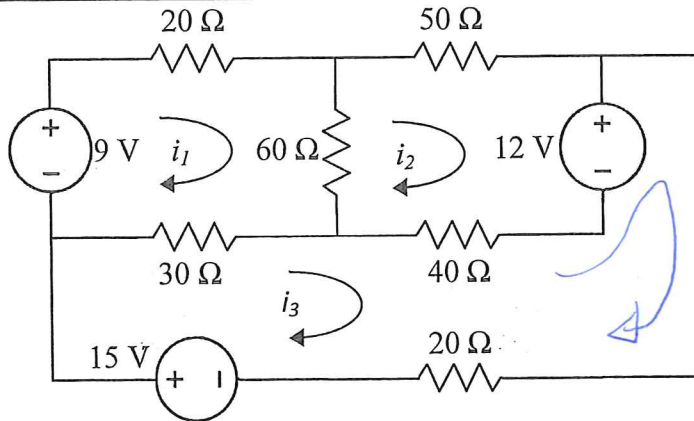
[30 min / 15 marks]

Question 1 [3]

Determine the mesh-current equations by means of inspection.

Vraag 1 [3]

Bepaal die lus-stroom vergelykings deur inspeksie.



Question 2 [4]

Given: $\beta = 99$ and $V_{BE} = 0.7$ V.

a) If $I_C = 0.5$ mA, determine V_S . [2]

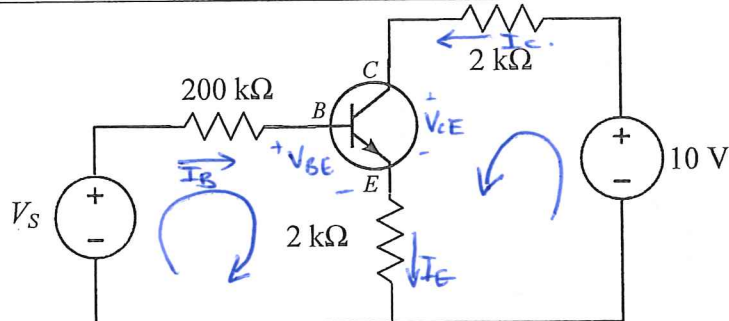
b) If $I_B = 10$ μ A, determine V_{CE} . [2]

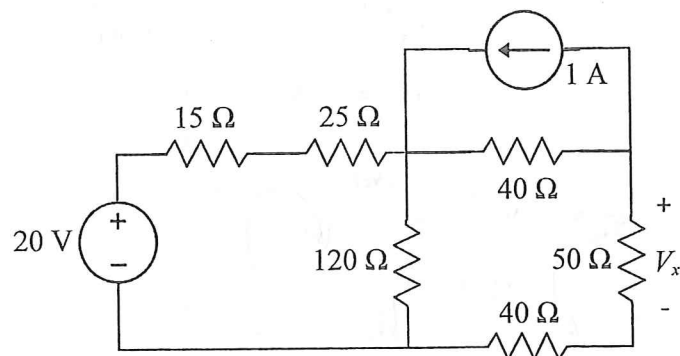
Vraag 2 [4]

Gegee: $\beta = 99$ en $V_{BE} = 0.7$ V.

a) Indien $I_C = 0.5$ mA, bepaal V_S . [2]

b) Indien $I_B = 10$ μ A, bepaal V_{CE} . [2]



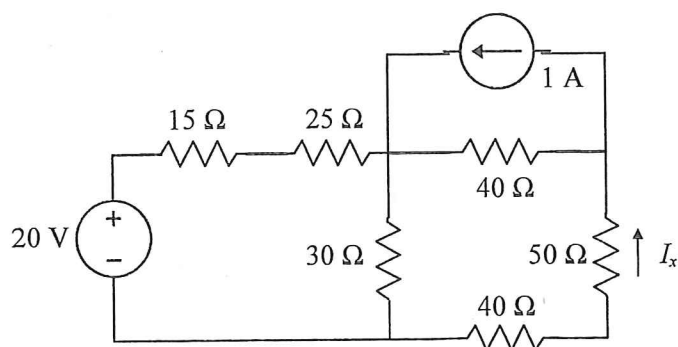
Question 3 [2]Use source transformation and determine V_x .**Vraag 3 [2]**Gebruik brontransformatie en bepaal V_x .

Question 4 [2]

Use superposition and determine I_x due to the current source.

Vraag 4 [2]

Gebruik superposisie en bereken I_x agv die stroombron.



Question 5 [4]

Gebruik superlus-analyse om twee vergelykings van die volgende vorm op te stel:

$$ai_1 + bi_2 = 15$$

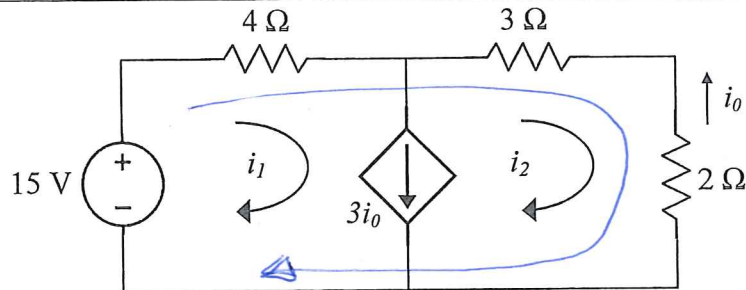
$$ci_1 + di_2 = 0$$

Vraag 5 [4]

Use supermesh analysis to set up two equations of the following form:

$$ai_1 + bi_2 = 15$$

$$ci_1 + di_2 = 0$$



EBN111 CLASS TEST 2 / KLASTOETS 2 (8 APR 2012)
ANSWER SHEET / ANTWOORDBLAD

<i>Surname:</i> Van:		<i>Student number:</i> Studentennummer:						
<i>Initials:</i> Voorletters:		<i>Group:</i> Groep:	ENG		ENG (Mech)		AFR	

Tot:	/15
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1	$\begin{bmatrix} \text{[redacted]} \\ \text{[redacted]} \\ \text{[redacted]} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} \text{[redacted]} \end{bmatrix}$ ✓✓✓	
2	a) $V_S =$ [redacted] b) $V_{CE} =$ [redacted]	
3	$V_x =$ [redacted]	
4	$I_x =$ [redacted]	
5	a = [redacted]	b = [redacted]
	c = [redacted]	d = [redacted]

